



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

BOOKLET A

NAME

INDEX NO

CG

BIOLOGY

9744/02

Paper 2 Structured Questions

1 Sep 2021

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all questions in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, **submit booklets A, B and C separately** to the invigilator.

For Examiner's Use	
Section A	
1	14
2	10
3	12
4	11
5	12
6	12
7	14
8	10
9	5
Total	100

This document consists of **25** printed pages and **3** blank pages.

Answer **all** questions.

- 1 (a)** Describe the features of embryonic stem cells.

[3]

- (b)** Fig. 1.1 shows the changes in energy during the progress of an uncatalysed reaction.

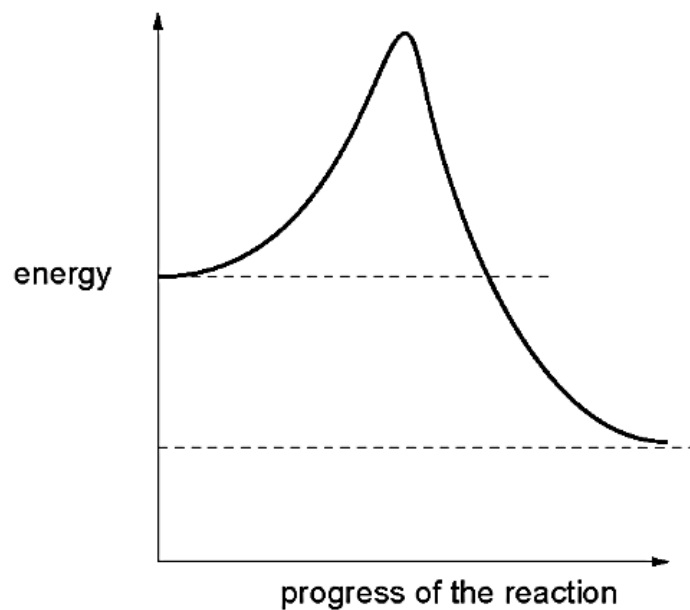


Fig. 1.1

- (i)** Draw on Fig. 1.1, a curve to show changes in energy during the progress of the same reaction when catalysed by an enzyme. [2]
- (ii)** State the term given to the energy level that must be overcome before a reaction can progress. [1]

- (c) Sox2, Oct4 and Nanog are transcription factors involved in maintaining the potency of stem cells. When E3 ligase acts on them, differentiation of stem cells occur.

Some of the key events of differentiation of stem cells are shown in Fig. 1.2

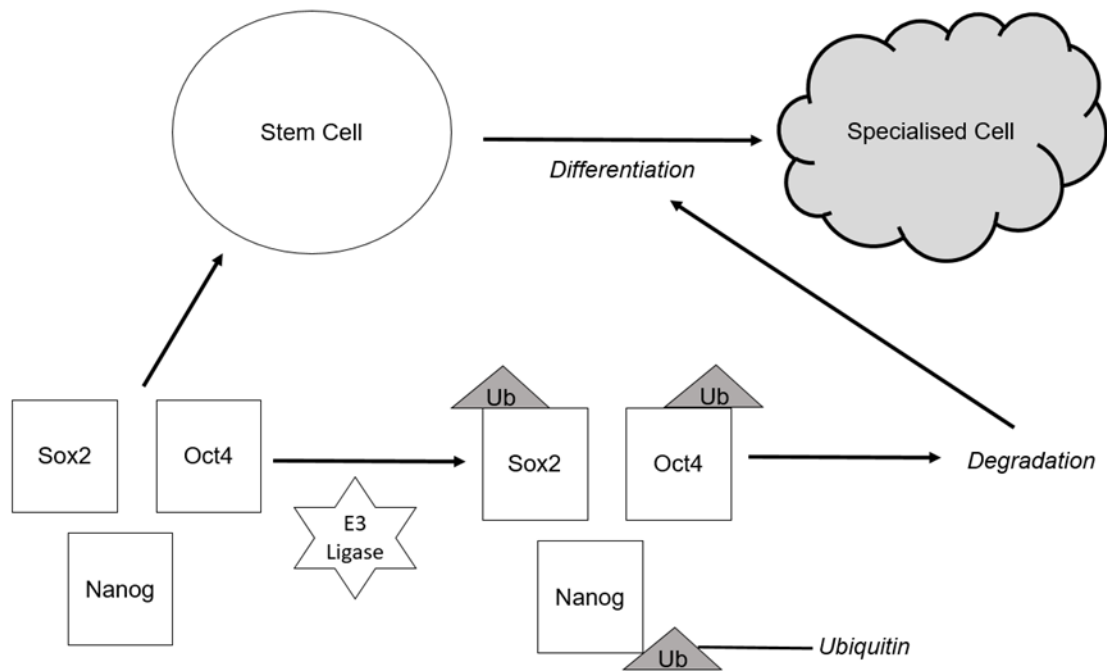


Fig. 1.2

With reference to the information and Fig. 1.2 above, describe the function of E3 ligase in the differentiation of stem cells.

[2]

Fig. 1.3 shows the effect of increasing protein substrate concentration on the rate of ubiquitination. All other variables were controlled.

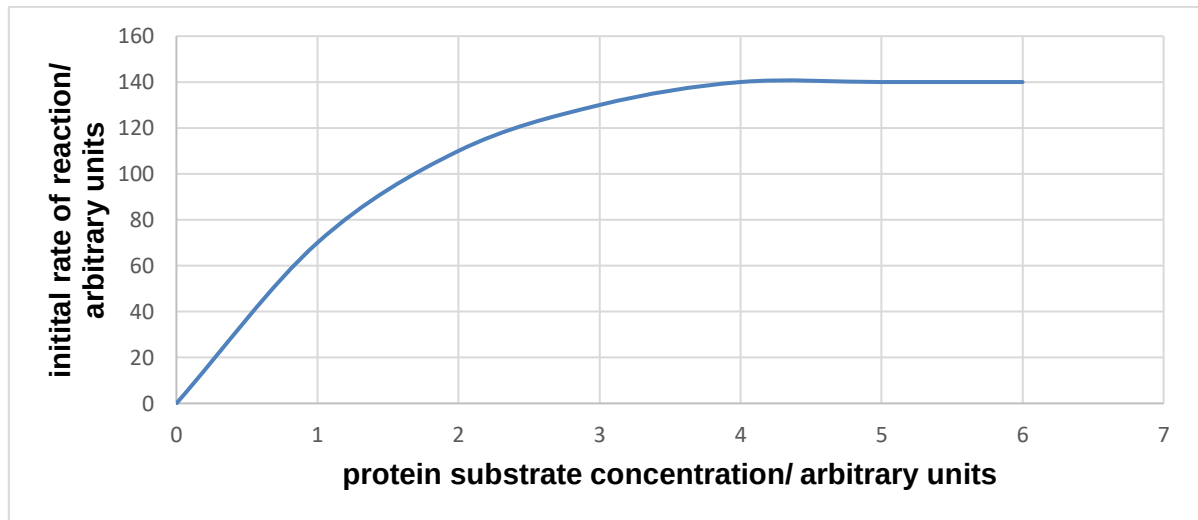


Fig. 1.3

(d) With reference to Fig. 1.3,

(i) describe the effect of increasing protein substrate concentration on activity of E3 ligase.

[2]

(ii) explain your answer in **(d)(i)**.

[3]

(e) State the limiting factor where the graph plateaus in Fig. 1.3.

[1]

[Total: 14]

2 Fig. 3.1 shows a segment of a DNA molecule.

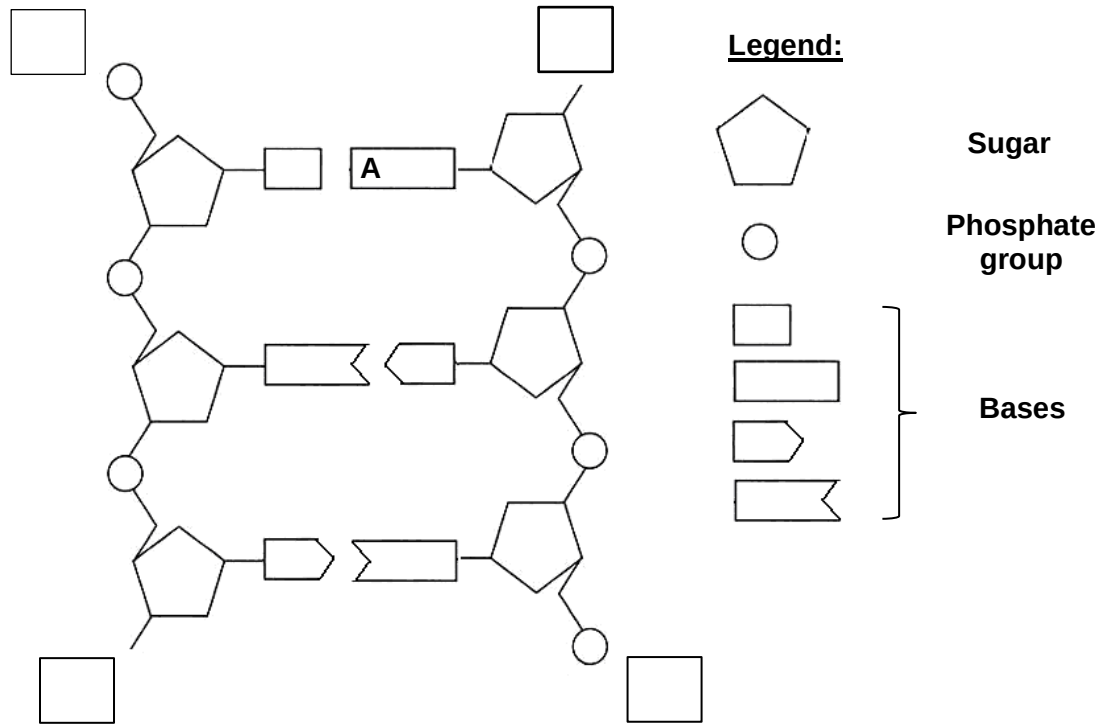
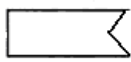


Fig. 2.1

(a) Identify the following bases:



[2]

(b) State the type of pentose sugar present in this molecule.

[1]

(c) On Fig. 2.1, label the 5' and 3' ends on **both** polynucleotide strands.

[1]

(d) Describe **two** structural differences between DNA and RNA.

[2]

- (e) The DNA molecule is synthesised via a process called DNA replication.

Explain the role of DNA polymerase III in this process.

[2]

- (f) Fig. 2.2 shows a bacterial chromosome and a human chromosome.

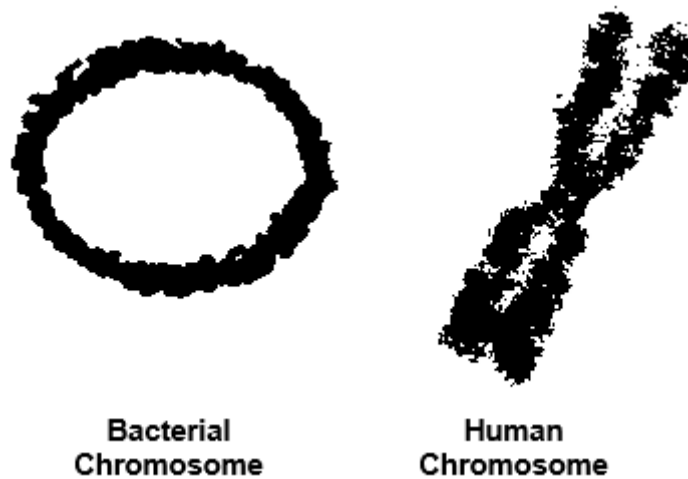


Fig. 2.2

With reference to Fig. 2.2, suggest why end-replication problem does not occur in the bacterial chromosome.

[2]

[Total: 10]

3 Fig. 3.1 shows the changes in DNA content in a cell during the cell cycle.

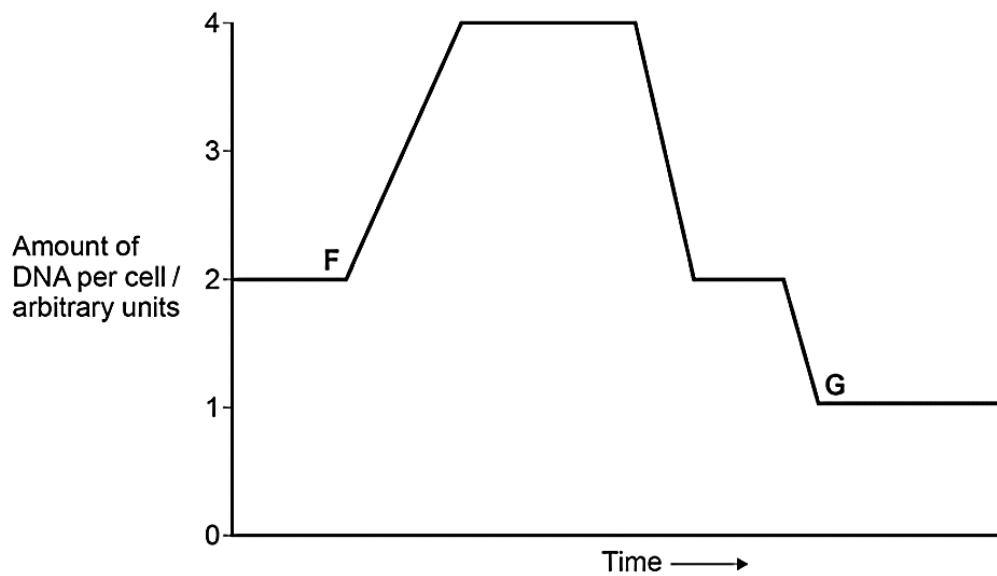


Fig. 3.1

(a) With reference to Fig. 3.1,

(i) state the type of nuclear division shown.

[1]

(ii) explain how the behaviour of chromosomes causes these changes in the amount of DNA per cell between F and G.

[3]

(iii) explain the significance of the process in (a)(i).

[2]

Microtubules are made from polymerisation of tubulin dimers. Fig. 3.2 shows the structure of the tubulin dimer.

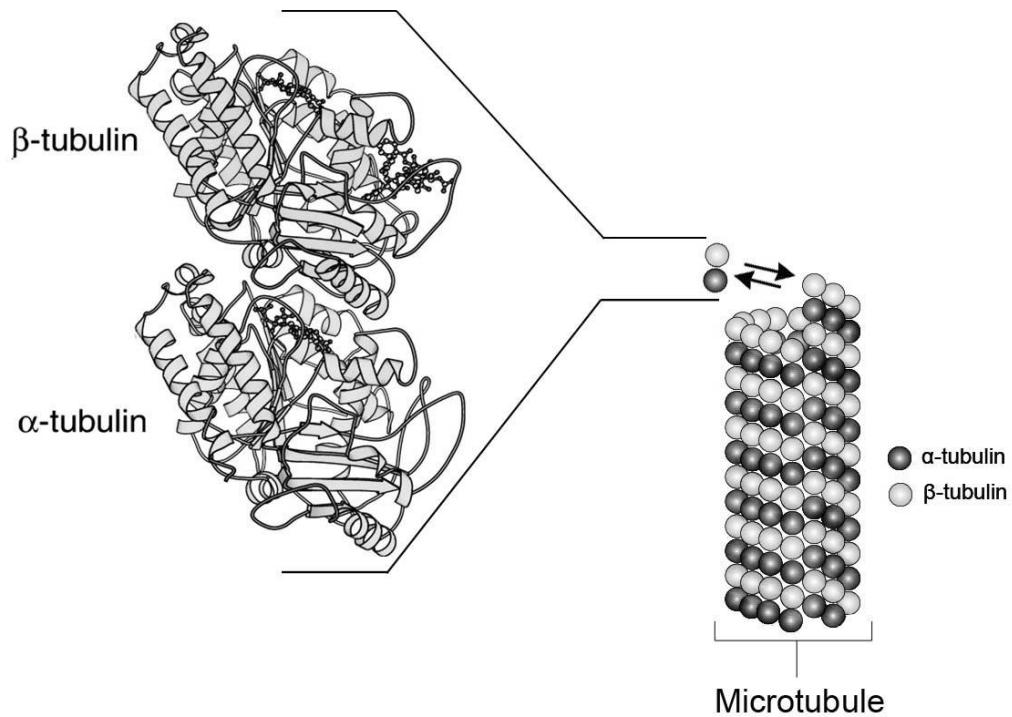


Fig. 3.2

- (b)** With reference to Fig. 3.2 and your knowledge of level of organisation of protein structure, describe the structure of tubulin.

[3]

- (c) Tubulin inhibitors like paclitaxel and vinblastine have been utilised in chemotherapy drug trials to treat cancers. Tubulin inhibitors are known to bind to the β -tubulin subunit.

Suggest how tubulin inhibitors reduce tumour formation.

[3]

[Total: 12]

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- 4 *Streptococcus pneumoniae* is a bacterium normally present in the nose and throat of healthy individuals. However, it can cause serious diseases such as pneumonia (infection of lungs) and bacterial meningitis (infection of the membrane around the brain).

- (a) Describe how the bacterial DNA is packaged to fit into the nucleoid region of a *S. pneumoniae* cell.

[2]

- (b) Antibiotic resistance in *S. pneumoniae* is a major concern for clinicians.

Explain how an antibiotic resistant strain of *S. pneumoniae* could have transferred the antibiotic resistance gene to normal *S. pneumoniae* that is present in the throat of a healthy adult.

[3]

- (c) To eradicate the harmful bacteria, some scientists researched on developing a modified bacteriophage that could destroy the bacteria.

Describe and explain how the reproductive cycle of such bacteriophages would be able to destroy the bacteria without harming humans.

[4]

- (d) A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes of the *lac* operon in *S. pneumoniae*, to give prophages.

Explain the change in number of bacteriophage, if any, when host bacteria are grown in environment with an abundant supply of glucose and no lactose.

[2]

[Total: 11]

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- 5** Wiskott-Aldrich syndrome is an example of rare disorders in which there is a defect in one or more aspects of the immune system. In Wiskott-Aldrich syndrome, the functioning of B lymphocytes and T lymphocytes is defective and there is a low platelet count, with the size of platelets being smaller than normal.

- (a)** A study of six generations of one family led to the conclusion that the Wiskott-Aldrich syndrome showed X-linked inheritance.

Grace who is normal, has a father who is normal and a brother who died of Wiskott-Aldrich syndrome when he was three years old. She marries Joshua who does not have Wiskott-Aldrich syndrome. The couple decides to have a child.

It was later discovered that both are heterozygous for the gene coding for the enzyme tyrosinase which catalyses the first two steps in the conversion of tyrosine to melanin, the major pigment found in melanocytes. Albinism is an autosomal recessive disorder which affects the production of melanin.

Using the following symbols,

X^A: dominant allele for no Wiskott-Aldrich syndrome on X chromosome

X^a: recessive allele for Wiskott-Aldrich syndrome on X chromosome

N: dominant allele for normal melanin production

n: recessive allele for albinism

draw a genetic diagram, explain the probability that their first son has both Wiskott-Aldrich syndrome and albinism.

Probability that 1st child is an albino male with Wiskott-Aldrich syndrome: _____ [5]

In humans, two genes control the presence of a rare form of diabetes mellitus. These two genes, **R/r** and **E/e**, code for the insulin receptor tyrosine kinase (RTK) and insulin receptor substrate 1 (IRS1) respectively.

In high blood glucose concentrations, insulin binds to RTK, initiating a series of responses that result in the activation of IRS1. Activated IRS1 then phosphorylates other relay proteins, transducing the signal into a variety of cellular responses.

In a case study, the phenotypes of offspring from double heterozygotic parents were traced and recorded.

normal	91
diabetic	68

(b) (i) Identify the type of interaction between the gene loci

[1]

(ii) Explain how you arrived at your answer in part **(i)**.

[3]

From the data observed, a geneticist hypothesised that inheritance of the rare form of diabetes mellitus follows the expected phenotypic ratio of 9 normal: 7 diabetic.

A χ^2 test was carried out and a calculated value of 0.0624 was obtained.

Part of the critical values of the chi-squared distribution is shown in Table 5.1 below.

Table 5.1

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (c) Using Table 5.1 and the calculated χ^2 value, explain the conclusion that can be drawn regarding the geneticist's hypothesis.

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[3]

[Total: 12]

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6 Fig. 6.1 shows a process occurring during aerobic respiration.

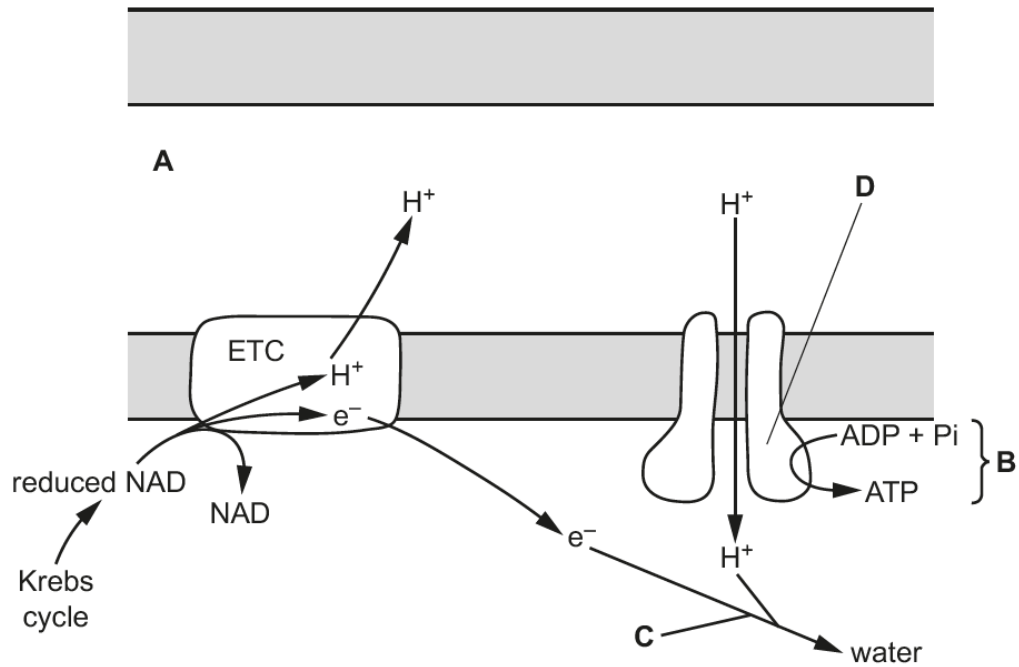


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) describe the role of molecule **C** in accumulation of protons at region **A**.

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.....

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.....

.....

[3]

(ii) describe the role of molecule **D** in process **B**.

.....

.....

.....

.....

[2]

- (b) During anaerobic respiration, molecule **C** is absent.

Briefly describe how muscular contractions in an athlete can continue in the absence of molecule **C**.

[2]

An experiment was conducted to investigate respiration in small aerobic crustaceans. Fig. 6.2 shows the two experimental set-ups, **P** and **Q**.

The distance moved by the coloured liquid marker in the capillary tube was used to determine the volume of gases exchanged during respiration. Soda-lime, which absorbs carbon dioxide, was added to set-up **P**.

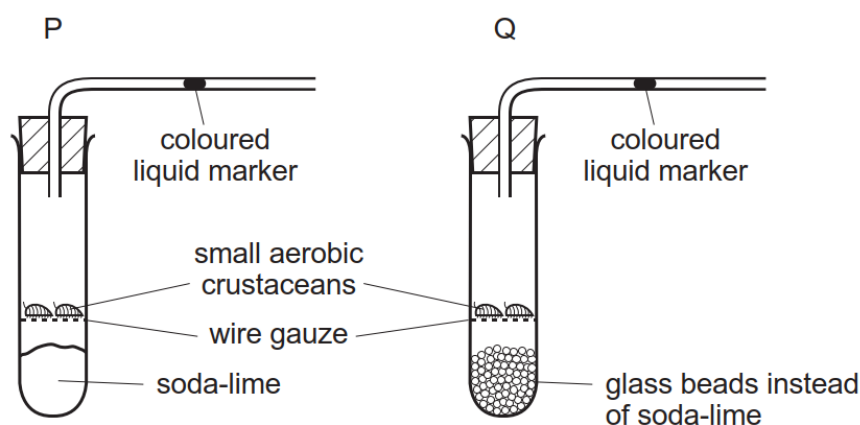


Fig. 6.2

Table 6.1 shows the results obtained.

Table 6.1

set-up	distance moved by coloured liquid marker / mm
P	8
Q	3

- (c) (i) On Fig. 6.2, use an arrow to indicate the direction of movement of the coloured liquid marker in set-up **P**. [1]

- (ii) Using your knowledge of the processes involved in aerobic respiration, explain your answer in (c)(i).

[2]

- (iii) The respiratory quotient (RQ) is the ratio of the volume of carbon dioxide produced to the volume of oxygen consumed by the body per unit time. It can be used to give an indication of the respiratory substrate used by an organism.

$$\text{RQ} = \frac{\text{volume of carbon dioxide produced}}{\text{volume of oxygen consumed}}$$

Using the information from Table 6.1, as well as the following information, calculate the respiratory quotient.

diameter of capillary tube = 0.4 mm

area of circle = πr^2

$\pi = 3.14$

[2]

[Total: 12]

Name: _____ ()

CG: _____

- 7 A subspecies is a genetically distinct population within a species that has some phenotypic differences from the rest of the species, but is not yet reproductively isolated.

Nine subspecies of the tiger, *Panthera tigris*, have been identified. Six of these subspecies are found on mainland Asia. Three of the subspecies originate from the Sunda Islands. These islands include Bali, Java and the large island of Sumatra.

Fig. 7.1 shows these three islands.

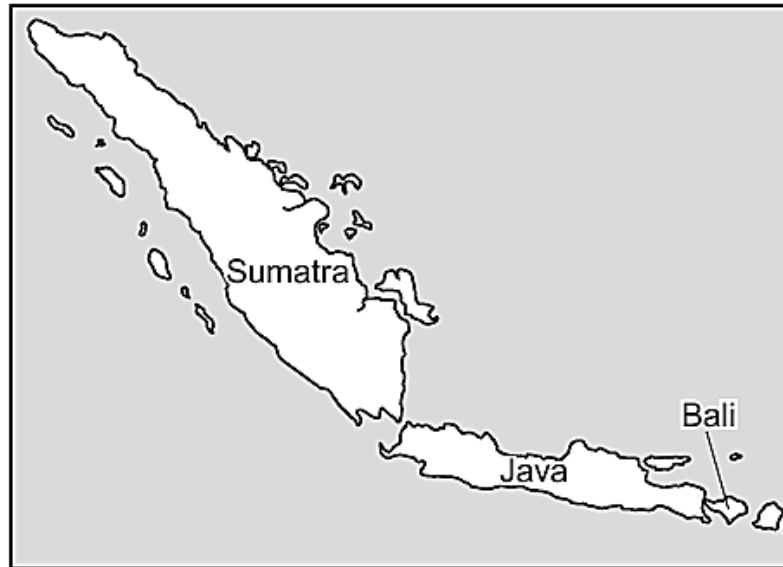


Fig. 7.1

- The Bali tiger, *Panthera tigris balica* (*P. t. balica*), became extinct in the 20th Century. The Bali tiger was found only on the island of Bali.
- The Javan tiger, *P. t. sondaica*, became extinct in the 20th Century. The Javan tiger was found only on the island of Java.
- The Sumatran tiger, *P. t. sumatrae*, lives only on Sumatra and is the closest living relative of Bali and Javan tigers.

20 000 years ago land bridges temporarily connected the Sunda Islands.

- (a)** Suggest **and** explain how the three subspecies of tiger on the Sunda Islands formed.

[5]

A recent study carried out a genetic analysis of the nine subspecies of tiger. Specific sections of mitochondrial DNA (mtDNA) were amplified using PCR and compared to assess their evolutionary history.

- The source of DNA for the extinct subspecies came from museum specimens.
- The mtDNA sections for the three island subspecies were genetically distinct from the other six mainland subspecies.

- (b)** Describe **one** characteristic of mtDNA that makes it more useful than using nuclear DNA to provide evidence of evolution.

[1]

- (c) Different pairs of primers are used to amplify different regions of the mtDNA from the nine subspecies.

Explain why a specific pair of primer is used for amplification of a specific region of the mtDNA. .

[1

- (d) Suggest why *P. t. sumatrae* is still considered to be a member of the species *Panthera tigris*.

[2]

- (e) State the genus of the Bali tiger.

[1]

Fig. 7.2 shows the phylogenetic tree of big cats constructed based on nucleotide sequences obtained from fossil and biogeographical data.

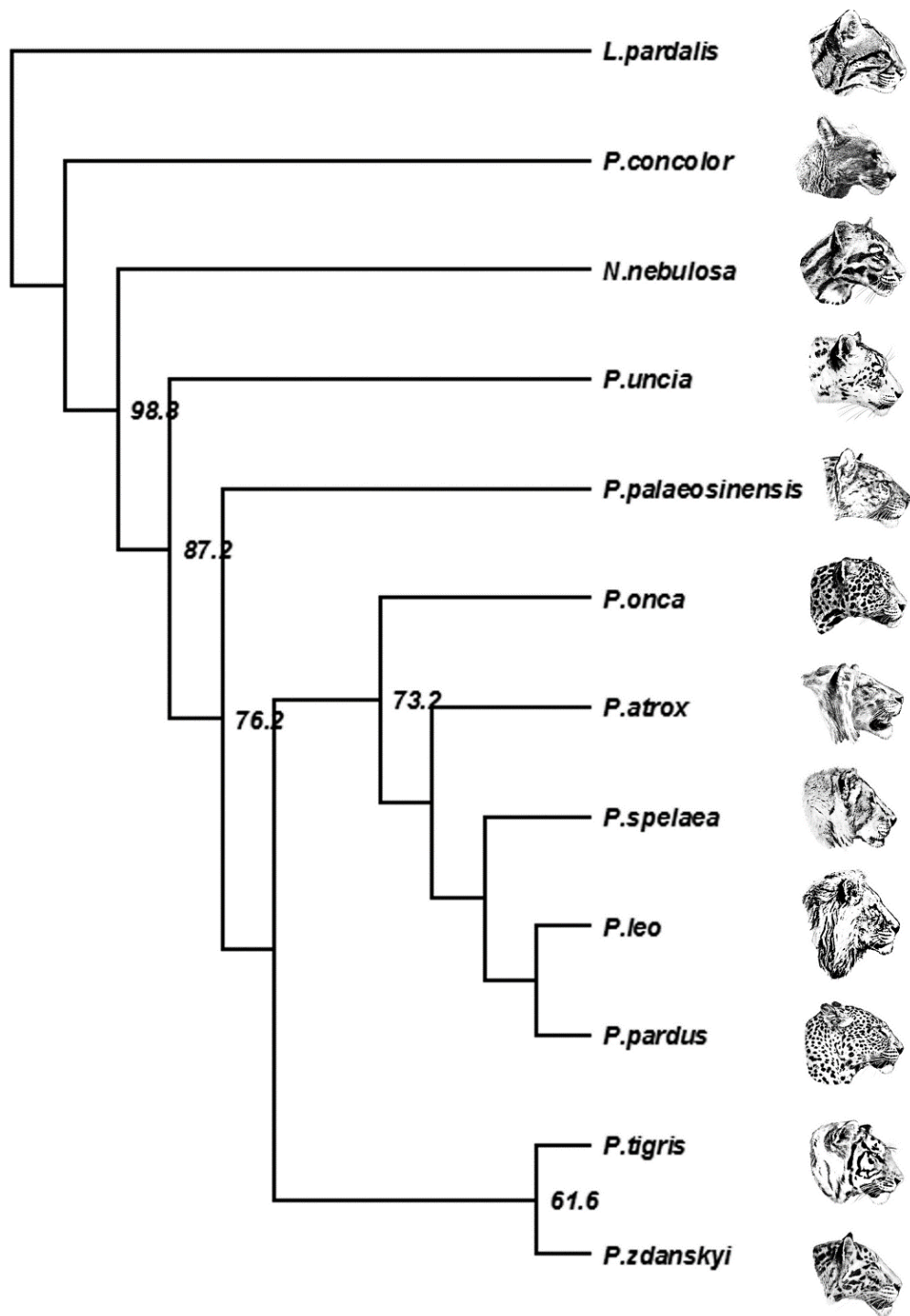


Fig. 7.2

- (f) (i) Explain how nucleotide sequences can be used in reconstructing phylogenetic relationships.

[2]

- (ii) Describe **one** advantage of using method in (f)(i).

[1]

- (iii) Label with an 'X' in Fig. 7.2, the most recent common ancestor of *P. spelaea* and *P. leo*.

[1]

[Total: 14]

8 Cry toxins are present in insecticides.

One way in which cry toxin causes cell death in target insects is by oligomerising into pore complexes in cell surface membrane of mid-gut epithelial cells.

- (a) Explain how formation of pores in the cell surface membrane of mid-gut epithelial cells causes death in target insects.

[2]

- (b) Another mechanism of action of Cry toxin in mid-gut epithelial cells of target insects has been proposed. Fig. 8.1 shows the proposed magnesium ion (Mg^{2+})-dependent signalling pathway.

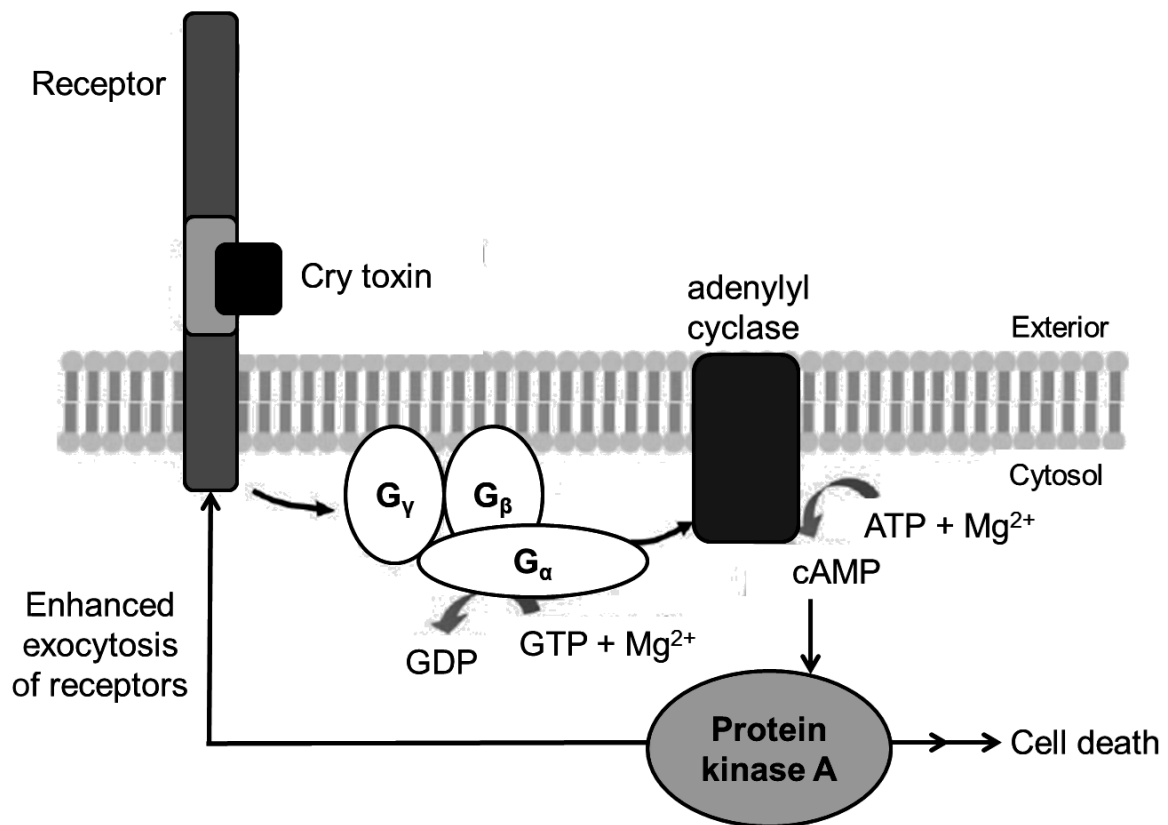


Fig. 8.1

- (i) With reference to Fig. 8.1, describe the proposed cell signalling pathway of Cry toxin.

[5]

- (ii)** NF449 is a drug that prevents binding of GTP to G protein.
- Suggest how NF449 could affect the signalling pathway of Cry toxin.

[2]

- (c)** Suggest why Cry toxins only kill target insects and not other organisms.

[1]

[Total: 10]

- 9 Whooping cough is a disease that affects some infants. Doctors collected data relating to whooping cough from 1965 to 1995.

They collected data for:

- the number of cases of whooping cough reported
- the percentage of infants vaccinated against whooping cough.

Fig. 9.1 shows the data collected by the doctors.

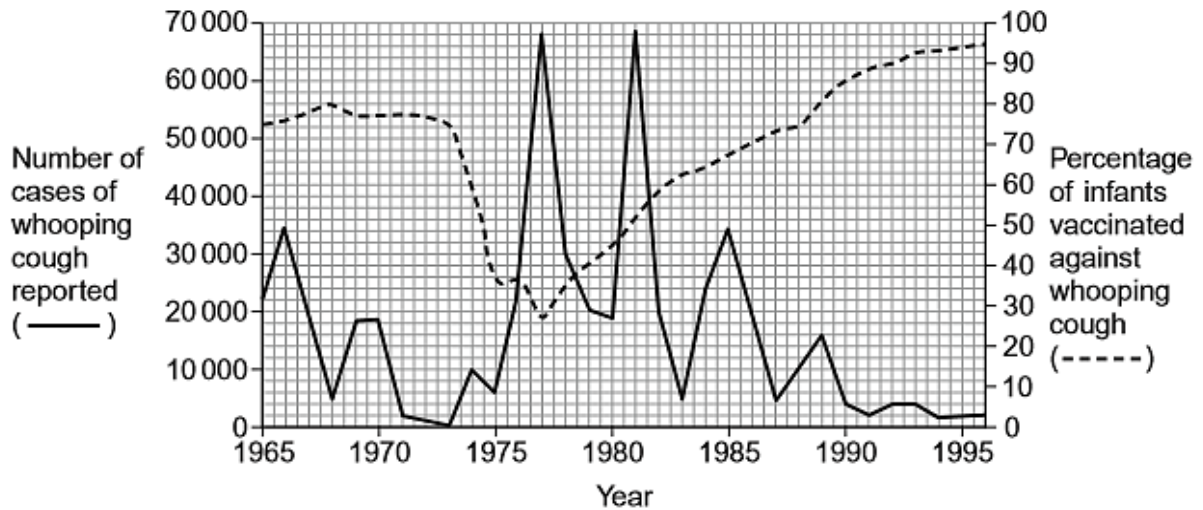


Fig. 9.1

(a) Using information from Fig. 9.1, suggest

- (i) why the percentage of infants vaccinated decreased between 1973 and 1975.

[2]

- (ii) why each successive peak in the number of reported cases of whooping cough decreased between 1980 and 1990.

[3]

[Total: 5]